

2023-Ph-H1-Q4

Section: Particles and Waves

Topic: Wave-Particle Duality

What provides evidence of wave-like behaviour in electrons?

Answer: A

(Diffraction)

Guidance for Students:

Electrons create diffraction patterns — like light — when passed through narrow slits. This is direct evidence that they can behave as waves.

Revision Tips:

- Electron diffraction = wave-like property.
- de Broglie wavelength: $\lambda = \frac{h}{p}$
- Requires low energy electrons and thin targets (e.g. graphite).